Capstone Assignment: Mini-Courier Planner

Lab Overview

Mapped Course Outcome: CO1, CO2, CO3, CO4, CO5 (L5 - Evaluate)

Total Marks: 100

Focus Areas: All topics covered above

Problem Statement

Build a prototype that, given a list of parcels and delivery points, generates an ordered delivery route and an optimized packing plan. The solution should integrate sorting, dynamic programming, graph algorithms, and approximation techniques to simulate an intelligent courier planning system.

Capstone Instructions

Develop an integrated Python application that combines the algorithms implemented in previous lab assignments. The system should prioritize parcels, optimize package selection, determine the shortest delivery routes, and compare the runtime performance of different algorithmic components. Submit the complete project through file upload or an accessible GitHub repository with source code, documentation, sample datasets, screenshots, runtime charts, and execution steps.

Learning Outcomes

After successfully completing this Capstone Project, students will be able to:

- Integrate multiple algorithmic paradigms, including sorting, dynamic programming, graph algorithms, and approximation techniques, to develop a complete solution for a real-world computational problem.
- Design and implement a modular Python application by selecting appropriate data structures and algorithms based on problem requirements.
- Evaluate the performance of different algorithmic components using execution time, computational complexity, and scalability analysis.
- Justify the selection of algorithms by comparing alternative approaches and demonstrating their suitability for the given problem.
- Develop, test, and document a complete software solution using appropriate datasets,

experimental observations, and technical reports.

- Demonstrate problem-solving, critical thinking, and software engineering skills through project implementation, presentation, and technical discussion.

Capstone Assignment: Mini-Courier Planner				
Submission Mode	File upload or GitHub repository link			
Phases of Capstone Assignment - 100 Marks				
Phase	**Meaningful Title**	**Task to be Completed**	**Evidence / Submission**	**Marks**
Phase 1	Courier Problem Design and Planning	Design the courier-planning problem, define parcel attributes, delivery points, graph format and required algorithms. Prepare a simple design showing how sorting, knapsack, shortest path and route planning will work together.	Problem clarity, algorithm selection, data design, route-planning logic and suitability of selected techniques.	20
Phase 2	Package Prioritization and Box Filling	Sort parcels based on delivery deadlines or priority using the best sorting algorithm. Apply the Dynamic Programming to determine the optimal set of parcels for one delivery trip.	Correct parcel sorting, DP table or output evidence, selected parcel list and packing-plan result.	20
Phase 3	Route Maker and Graph Processing	Build the customer graph, use Dijkstra's Algorithm for individual shortest paths and apply nearest-neighbor TSP approximation for the overall delivery route.	Graph representation, shortest-path output, final route sequence and route-length calculation.	20
Phase 4	Performance Testing and Optimization	Test the system using small and larger datasets. Record runtime of major steps and suggest improvements for future optimization.	Test cases, runtime chart, correctness evidence, optimization comments and comparison of major steps.	20
Phase 5	Final Report and Demonstration	Present the working system with screenshots or output traces, explain the result and submit the final file or GitHub repository with proper documentation.	Result explanation, documentation quality, screenshot evidence, final demonstration and originality.	20
Total				**100**

Submission Checklist

- Complete, well-commented, modular, and executable Python source code for the integrated **Mini-Courier Planner**project.
- Proper project structure containing source code, datasets, screenshots, runtime charts, and supporting documentation.
- Sample parcel dataset and weighted graph dataset used for testing the application.
- Console outputs and screenshots demonstrating parcel prioritization, capacity optimization, route planning, and final results.
- Runtime comparison tables and charts showing the performance of major project modules for different input sizes.
- Complexity analysis and performance observations for the implemented algorithms.
- A well-structured project report including problem statement, system design, methodology, results, conclusion, and future scope.
- A **README.md** file containing project overview, setup instructions, execution steps, dependencies, and repository structure.
- GitHub repository link (or file upload) submitted through the LMS/ZOSIMA portal before the deadline.
- Original work with proper documentation and readiness for project demonstration and viva as per faculty instructions.

Push the following files to your private GitHub repository:

- **README.md** – Project overview, installation and usage
- **mini_courier_planner.ipynb** – Complete implementation
- **src/** – Python source files (optional modular implementation)
- **datasets/** – Sample parcel and route datasets
- **images/** – Runtime graphs, route visualizations and screenshots
- **requirements.txt**
- **.gitignore**

Tag the final commit as:

v1.0-final-submission

Submit the GitHub repository URL through the LMS/Zosima.